

Mid-ordinate rule

Key Points

To approximate the integral

$$\int_a^b f(x) dx$$

Split the interval from a to b into n strips each of width $h = \frac{b-a}{n}$. Set

$$x_0 = a, \quad x_i = x_{i-1} + h,$$

$$x_{i+0.5} = \frac{x_i + x_{i+1}}{2}$$

Set $y_{i+0.5} = f(x_{i+0.5})$. Then

$$\int_a^b f(x) dx \approx h (y_{0.5} + y_{1.5} + \dots + y_{n-0.5})$$

Basic Skills

Q1. Calculate the width of the strips when we split the interval 0 to 10 into

- a) 10 strips b) 5 strips c) 4 strips

Q2. If $h = 2$ and $x_0 = 5$, what are

- a) x_1 b) x_2 c) x_3 d) x_5

Q3. If $f(x) = x^2$, find $y_{0.5}$ if $x_{0.5} = 2.5$

Q4. Using the following table calculate $\int_1^3 f(x) dx$

$x_{i+0.5}$	1.25	1.75	2.25	2.75
$y_{i+0.5}$	6	3.5	2	1

Competency Skills

Q5. We want to approximate $\int_0^5 e^x dx$ using the mid-ordinate rule using 5 strips

a) Fill in the table

b) Using the Mid-ordinate rule, approximate $\int_0^5 e^x dx$

c) Determine if the Mid-ordinate rule is an over or under estimate

Q6. Approximate $\int_0^5 e^{x^2} dx$ using the mid-ordinate rule with 4 strips

Q7. Approximate $\int_{-\pi}^{\pi} \sin^4 x dx$ using the mid-ordinate rule with 6 strips stating how else this estimation could be done

Q8. Use the mid-ordinate rule with 8 strips to find an approximate to $\int_0^5 e^x dx$ and compare your answer to that in Q5.

i	0	1	2	3	4
$x_{i+0.5}$					
$y_{i+0.5}$					

Advanced Skills

Q9. One of your friends has a maths problem and asks you for help. They need to roughly calculate $\int_1^3 \ln(x) dx$ but they don't know how to integrate $\ln(x)$ (as they haven't learnt integration by parts). Can you help them

Q10. State a function for which the mid-ordinate rule calculates the exact area

Q11. The function $f(x) = [x]$ has the effect of rounding x down to the nearest whole number. You are asked to calculate $\int_0^2 [x] dx$. Since you don't know how to integrate this function your friend now suggests using the midpoint method with 4 strips. Your friend has done more work than they needed to. Why?

Extension

Q12. Calculate $\int_0^1 x^2 dx$ using n strips, for some positive whole number n . You may use the following formula for the sum of the first n odd numbers squared: $1^2 + 3^2 + \dots + (2n-1)^2 = \frac{1}{3}(n(2n+1)(2n-1))$. What happens as n gets bigger?